

AI cost, profit & token audit

Checked 28 September 2026 against the live code and the providers' official price pages. All ₹ figures use ₹95.93 per US\$ (ExchangeRate-API, 28 Sep 2026).

At a glance

- We keep **about 86%** of every ₹50 single shot and **about 92%** of every ₹500 pack after paying for the AI.
- The finished image is ~97% of the AI bill. Our written instructions (prompts) cost about ₹0.10 per image.
- **Urgent:** the daily image limit is set to 1, so every pack from a customer outside our team is refused and auto-refunded right now.
- **Urgent:** the backup image maker (OpenAI gpt-image-1) shuts down on **23 October 2026**.
- The pack in the code has **6 shots, not 8**. Both are shown below.

Part A — What our AI costs and what we keep

A1. Which AI actually runs our system

Job	AI used	In plain words
Photo check	Google Gemini 3.6 Flash	Looks at every photo a customer sends: is it one clear piece of jewellery, not blurry? Runs on every photo, even if no order follows. If the check itself breaks, the photo is let through.
Photo maker	Google Gemini 3.1 Flash Image (“Nano Banana 2”)	Takes the customer's photo plus our written instructions and makes one new product photo. Standard size (1K), one attempt.
Backup photo maker	OpenAI gpt-image-1, high quality	Used only when Google fails with certain errors. Costs 3–5× more than Google. Shuts down 23 Oct 2026.
Claude (Anthropic)	Not used	Only an unfinished placeholder file that nothing connects to. Claude can read photos and write text but cannot create images, so at most it could do the photo check.

Why Google? It is the cheapest good-quality AI that can edit from the customer’s own photo: about ₹6.4 per image, against ₹16–24 on OpenAI.

A2. What each customer photo costs us

Item	Cost	Share of the bill
Photo check	₹0.28 per photo (up to ₹0.83); about ₹0.55 from 1 Jan 2027 when Google's price doubles	Tiny
Making one finished image	₹6.43 at 1K (₹9.69 at 2K)	About 97%
Our written instructions (prompt)	About ₹0.10 per image	1–2%

Google charges only for requests that succeed. Tokens (small chunks of text the AI counts) are roughly 4 characters, or ¾ of an English word, on all three providers; Part B measures our real ratio.

A3. Margin breakdown (Google, standard 1K size)

Order	We charge	Actual AI cost	Net profit	Margin	Profit after 18% GST	Margin after GST
Single Shot	₹50	₹6.93	₹43.07	86.1%	₹35.44	83.6%
8-Shot Pack (as asked)	₹500	₹54.01	₹445.99	89.2%	₹369.72	87.3%
6-Shot Pack (what the code does)	₹500	₹40.58	₹459.42	91.9%	₹383.15	90.4%

Prices are GST-inclusive, so after GST we keep ₹42.37 of each ₹50 and ₹423.73 of each ₹500. At larger 2K images: single shot ₹10.19 (79.6% margin), 8-shot pack ₹80.10 (84.0%). **Not included:** Razorpay fees, card or currency-conversion charges, any GST on imported AI services, hosting.

A4. The biggest danger to our profit

Written instructions cost almost nothing — even very long ones add about ₹0.10 per image. **Every image that gets made is what costs money**, and mistakes and repeats make more images.

If every image used the OpenAI backup	AI cost	Profit	Margin
Single Shot	₹21.41	₹28.59	57%
6-Shot Pack	₹190.83	₹309	62%
8-Shot Pack	₹254.28	₹246	49%

- **Repeats:** remaking one image costs about ₹6.7. The manual “retry order” option re-runs the whole order, so one retry roughly doubles that order's cost.
- **Refunds:** a refunded order loses the whole ₹50 or ₹500, but images already made before the failure are still paid for.
- **Photos that never become orders:** each one is still checked, about ₹275 per 1,000 photos.

A5. Urgent actions

1. **Pick a real daily limit.** The daily AI image limit is 1. Every 6-shot pack from a non-team customer is refused and refunded before any AI runs, and only about one single shot a day goes through.
2. **Replace the backup before 23 Oct.** Switch to OpenAI's gpt-image-2 (roughly ₹22–26 per image) or remove the backup.
3. **Make retries cheaper.** A retry should remake only the images that failed, not the whole order. Track backup usage and refunds weekly — they decide whether we keep about 90% or fall towards 50%.

Part B — Tokens per word: our 7 live image prompts

Gemini vs Claude vs GPT-4o · measured on the production prompts (main @ c206a39)

Bottom line. Our prompts average **1.44 tokens per word on Gemini** and **1.37 on GPT-4o**, measured with each provider's own tool. The earlier 1.60–1.68 figure was not a measurement: it came from the rough “4 characters = 1 token” shortcut applied to our long words (6.68 characters per word). Claude is estimated: about **1.40** on older models and about **1.81** on the newest ones.

B1. Prompt-by-prompt comparison

Prompt	Chars	Words	Google Gemini (exact)		Claude ≤4.6 (estimate)		Claude 4.7+ (estimate)		OpenAI GPT-4o (exact)	
			Tokens	Per word	Tokens	Per word	Tokens	Per word	Tokens	Per word
P1 Clean Studio (₹50 white BG)	6,635	1,027	1,492	1.45	~1,471	1.43	~1,912	1.86	1,450	1.41
P2 Clean E-Commerce	9,746	1,452	2,092	1.44	~2,027	1.40	~2,635	1.81	1,990	1.37
P3 Close-up on Ear	8,898	1,397	1,887	1.35	~1,849	1.32	~2,404	1.72	1,827	1.31
P4 Scale Reference	15,137	2,285	3,314	1.45	~3,186	1.39	~4,142	1.81	3,143	1.38
P5 Professional Studio	12,375	1,753	2,531	1.44	~2,475	1.41	~3,218	1.84	2,405	1.37
P6 Lifestyle	14,025	2,065	3,055	1.48	~2,932	1.42	~3,812	1.85	2,871	1.39
P7 UGC Style	12,216	1,853	2,638	1.42	~2,576	1.39	~3,349	1.81	2,522	1.36
Total / average	79,032	11,832	17,009	1.44	~16,516	1.40	~21,471	1.81	16,208	1.37

How each column was produced. **Gemini:** exact, from Google's free token-counting service (gemini-3.1-flash-image and gemini-3.6-flash gave identical counts). **GPT-4o:** exact, from OpenAI's official tokenizer (o200k_base) run offline. **Claude ≤4.6** (Sonnet 4.6, Haiku 4.5): estimated with OpenAI's previous tokenizer (cl100k_base) as a stand-in, because no Anthropic key is available. **Claude 4.7+:** that estimate plus 30%, per Anthropic's statement that its newer tokenizer produces “approximately 30% more tokens”. **Words:** anything separated by spaces, so bullet dashes count as words. P1 is the ₹50 single shot; P2–P7 are the six E-Com Pack 1 shots.

B2. Why we are above the usual 1.30 rule of thumb

- The 1.30 rule describes everyday English, where most words are short and common, like “the” or “photo”. The AI stores those as a single token each.
- Our prompts read like a technical rulebook: long specialist words, CAPITAL headings and hundreds of bullet points. Those get split into more pieces, which puts us at about 1.4 tokens per word. That is only moderately above the rule.
- The 1.60–1.68 figure was a by-product of the shortcut: 6.68 characters per word ÷ 4 = 1.67. In reality Gemini fits about 4.65 of our characters into each token, so it uses fewer tokens than the shortcut predicted.
- Claude's newest models split text more finely and will land near 1.8 tokens per word on the same prompts.

B3. What formatting and jewellery terms add

Feature	What happens	Extra tokens (all 7 prompts)	Share
Bullet points and numbering (742 lines)	Each “- ” or “1.” marker, and each line break, costs a token that is not a real word	+656 markers, +331 line breaks	≈ 6%
CAPITALS (867 words)	Capitals are rarer, so they are split: “ABSOLUTE” = 3 tokens vs “absolute” = 1; “MUST” = 2	+425	2.6%
Hyphenated jewellery terms (201)	The hyphen breaks the word: “micro-pave” = micro -p ave (3); “lever-backs” = 3; “non-negotiable” = 4	+114	0.7%
Specialist / British spellings	“jewellery” = 3 tokens, “filigree” = 3, “earring” = 2	spread throughout	—
Hex codes (#FFFFFF, 11×)	“#FFFFFF” is only 2 tokens, but “RGB(255, 255, 255)” is 10 because every comma and number is its own piece	+5	≈ 0%
Other punctuation	Colons, brackets and dashes are usually 1 token each	+50	0.3%

Measured exactly on GPT-4o by removing each feature and recounting. Overall, about **24%** of our tokens are pure punctuation, spaces or numbers rather than words.

B4. What it means for cost

All of this formatting together adds roughly 1,500 tokens across the 7 prompts, about **₹0.07 in total** on Gemini (input text at \$0.50 per million tokens, ₹95.93 per US\$). The whole prompt text for a ₹50 single shot costs about ₹0.10; the finished image is about 97% of the AI bill. Trimming prompt formatting to save money is not worthwhile, and the bullets and capitals may help the AI follow the rules. Real cost control comes from limiting repeat generations and fallback use, not prompt length.

Exact Claude counts can be obtained free via Anthropic's token-counting endpoint once an API key is added.

Sources

Gemini API pricing — ai.google.dev/gemini-api/docs/pricing · Vertex AI pricing — cloud.google.com/vertex-ai/generative-ai/pricing · Gemini tokens, image generation and deprecations docs — ai.google.dev · OpenAI API pricing, image-generation guide and deprecations — developers.openai.com · OpenAI “What are tokens” — help.openai.com · Claude pricing and vision docs — platform.claude.com · Gemini count_tokens API; OpenAI tiktoken (o200k_base, cl100k_base) · FX: ExchangeRate-API (open.er-api.com), 28 Sep 2026; ECB rate via frankfurter.dev ₹95.82 (25 Sep). Code facts from [backend/app/config.py](#), [services/meta_whatsapp_service.py](#), [services/image_prevalidation_service.py](#), [ai/image_generation_manager.py](#) and [ai/providers/](#).